Qian Yang

Ph.D. Candidate, Mila / *Multimodal Agents: Reward Modeling and Reinforcement Learning*

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EDUCATION

Mila - Quebec AI Institute & Université de Montréal 2023.09 – 2027.11 (Expected)
Ph.D. in Computer Science | CGPA: 4.3/4.3 | Advisor: Prof. Aishwarya Agrawal
Research: Reliable Multimodal LLMs and Agents; Reward Modeling and RL

Harbin Institute of Technology, Shenzhen 2020.09 – 2023.03
M.Sc. in Computer Science and Technology | Advisor: Prof. Baotian Hu
Thesis: Fine-grained Alignment for Explainable Multi-modal Inference

University of Electronic Science and Technology of China (UESTC) 2016.09 – 2020.06
B.Eng. in Computer Science and Technology | CGPA: 3.73/4.0 (Top 10%)

RESEARCH SUMMARY

- **Reward Modeling & Verification for RL:** Built reward models that serve as RL training signals: a GRPO-trained rubric verifier deployed as the reward service for text-to-image RL, and preference reward models that capture multiple, style-dependent human preferences (**Tencent Hunyuan**); execution-based verification of LLM-generated code (**AAAI'25**).
- **Reasoning & Self-Verification:** Imagining unseen views as an intermediate reasoning state for spatial reasoning (**EMNLP'26**); self-verification by decomposing a question into sub-questions and checking consistency (**EMNLP'24**); multi-hop reasoning interleaved with retrieval (**ACM MM'23**).
- **Multi-Agent Collaboration:** Trained a question decomposer with **DPO** to collaborate with an answerer agent via **per-turn positive/negative pairs**: sub-questions that do vs. do not help the answerer reach the correct answer (**NeurIPS'24 Workshop**).
- **Adversarial Agents for Evaluation:** RL policies that learn to ask questions that make a target VLM fail, automatically discovering its failure modes (preprint); built the RL framework and baselines.

RESEARCH INTERNSHIPS

Tencent Hunyuan Team, China *(Qingyun Elite Internship Program)* 2026.06 – 2026.10
Reward Modeling for Text-to-Image Generation

- **Objective rubric reward model:** Trained a 27B VLM judge with **GRPO** (with a false-positive penalty) to give per-rubric 0/1 verdicts that are aggregated into a reward. Reached **89.1% per-rubric accuracy** with balanced errors (FP 10.9% / FN 10.4%) and 99.5% format validity; deployed on vLLM as the reward service for downstream T2I RL.
- **Subjective preference reward model:** Compared Bradley–Terry / Soft-BT, CoT vs. no-CoT, generative scoring, and distillation. Replaced a 27B generative judge with a **single 8B, no-CoT judge scoring in one forward pass**, reaching **83.7% accuracy on human-annotated preferences**.
- **Multi-preference reward modeling:** Found that **no single RM is optimal across visual-preference domains**: human preferences on technical vs. natural images agree only 65%, and an oracle RM ensemble beats the best single RM by 17–25pp. Proposed a **Style-Sensitive Reward Model** that automatically clusters preference styles and fits style-specific heads.

Alibaba DAMO Academy, Hangzhou, China 2022.05 – 2022.10
Multimodal Retrieval-Augmented Reasoning | Mentors: Dr. Wen Wang, Dr. Qian Chen

- Designed a **unified retrieval-generation** decoder that interleaves cross-modal retrieval with intermediate reasoning and **adaptively decides how many retrieval steps to take** for multi-hop question answering.
- Built entity-centric fusion over structured knowledge graphs to ground long-tail entities across images and text. Published at **ACM MM 2023**.

PUBLICATIONS

(* equal contribution)

Multimodal Reasoning & Verification

- **Qian Yang**, Ankur Sikarwar, ..., Aishwarya Agrawal. How and What to Imagine? Visual Thinking in Unified Multimodal Models for Cross-View Spatial Reasoning. **EMNLP 2026**.
  – Cross-view spatial reasoning, a core skill for embodied agents: a unified model **imagines an unseen view** as an intermediate reasoning step, and View Dropout makes the answer depend on this imagined state; best out-of-domain generalization on 5 real-world benchmarks.
- **Qian Yang**, Weixiang Yan, Aishwarya Agrawal. Decompose and Compare Consistency: Measuring VLMs' Answer Reliability via Task-Decomposition Consistency Comparison. **EMNLP 2024**.
  – **Self-verification via decomposition**: the model splits a question into sub-questions, answers them, and checks consistency with its direct answer, giving a test-time signal for when to trust its own output.
- Yuchen Tian, Weixiang Yan, **Qian Yang**, ..., Dawn Song. CodeHalu: Investigating Code Hallucinations in LLMs

via Execution-based Verification. **AAAI 2025**.

– Verifies LLM-generated code through **execution feedback**; 8.8K-sample benchmark exposing hallucinations across 17 LLMs.

- **Qian Yang**, Qian Chen, Wen Wang, Baotian Hu, Min Zhang. Enhancing Multi-modal and Multi-hop Question Answering via Structured Knowledge and Unified Retrieval-Generation. **ACM MM 2023**.

Efficient Training

- **Qian Yang***, Shivam Chandhok*, ..., Leonid Sigal, Aishwarya Agrawal. Learning What Matters: Prioritized Concept Learning via Relative Error-driven Sample Selection. **CVPR 2026**.

  – Relative-error-driven sample selection for VLM instruction tuning; **∼80% less data and compute**.

- Le Zhang, **Qian Yang**, Aishwarya Agrawal. Assessing and Learning Alignment of Unimodal Vision and Language Models. **CVPR 2025** (***Highlight***).

  – Aligns pretrained unimodal encoders to **match CLIP zero-shot with 6% of the paired data**.

Multi-Agent Collaboration & RL

- **Qian Yang**, Weixiang Yan, Aishwarya Agrawal. Enhancing Multi-Agent Multi-Modal Collaboration with Fine-Grained Reward Modeling. **NeurIPS 2024 Workshop** (AFM).

  – Two-agent system: a decomposer poses multi-turn sub-questions and an answerer answers them. The decomposer is **trained with DPO**: at each turn we sample several sub-questions, and those that **raise the answerer's confidence in the correct final answer** become positives, the rest negatives, turning a sparse trajectory-level signal into per-turn supervision for collaboration.

- Kanishk Jain, **Qian Yang**, ..., Aishwarya Agrawal. Discovering Failure Modes in Vision-Language Models using RL. *Under review*, 2026.

  – An RL policy learns to pose questions that make a target VLM fail, surfacing its failure modes automatically. *My role:* built the RL framework and the baselines (best-of-N, expert iteration, SFT).

Other

- **Qian Yang**, Yunxin Li, ..., Min Zhang. Chunk-aware Alignment and Lexical Constraint for Visual Entailment with Natural Language Explanations. **ACM MM 2022**.
- Le Zhang, Yihong Wu, **Qian Yang**, Jian-Yun Nie. Exploring the Best Practices of Query Expansion with Large Language Models. **EMNLP Findings 2024**.
- Yunxin Li, **Qian Yang**, ..., Lin Ma. Fast and Robust Online Handwritten Chinese Character Recognition with Deep Spatial & Contextual Information Fusion Network. **IEEE TMM 2022**.

AWARDS AND SCHOLARSHIPS

FRQNT Doctoral Scholarship, Fonds de recherche du Québec (25,000 CAD/year)	2026 – 2029
AI Scholarship, Université de Montréal (10,000 CAD)	2026
Women in Machine Learning (WiML) Scholarship, Mila (7,000 CAD)	2025
Student Success Scholarship, Université de Montréal (5,000 CAD)	2025
Professor Cho Diversity Scholarship, Mila (1,500 CAD)	2025
DIRO Excellence Scholarship, Université de Montréal (5,000 CAD)	2024 – 2025
National Encouragement Scholarship, China (Top 10%, national level)	2019
First Prize Academic Scholarship, UESTC (Top 20%)	2016 – 2020

ACADEMIC SERVICE & TEACHING

- **Workshop Organizer: VLMs4All** (Building Geo-Diverse and Culturally Aware VLMs), *CVPR 2025 Workshop*.
- **Reviewing: CVPR 2025 Outstanding Reviewer**; reviewer for CVPR 2024–26, ECCV 2024, AAAI 2024, ACM MM 2023–24, COLING 2022.
- **Teaching Assistant:** *Representation Learning* (IFT6135, graduate, Fall 2024, Université de Montréal); *Mathematical Logic*, *Intro to Algorithms* (2020–2021, HIT Shenzhen).

TECHNICAL SKILLS

- **Programming:** Python, C/C++, Bash
- **Training & Serving:** PyTorch, DeepSpeed (ZeRO-1/2/3), FSDP, FlashAttention, verl (Ray + vLLM), GRPO, DPO, Bradley–Terry reward modeling, LoRA / full-parameter SFT
- **Infrastructure:** multi-node SLURM clusters (A100/H100), Linux, Git, Docker, Weights & Biases
- **Languages:** English (fluent, TOEFL 99), Mandarin (native)